

Phenome-Wide Association Study of Hereditary Alpha-Tryptasemia in Epic Cosmos: Concise Report

For clinical and statistical review, 2026-10-08. A shortened version of the full status report (*HaT pheWAS preliminary report 10-7*), which has every table, the full list of 156 cluster phecodes, and the details behind each point below. Counts of 1–10 patients are masked (Cosmos small-cell rule).

1. What we did

Question. Which diagnoses, across the whole phenome, are more or less common in patients with hereditary alpha tryptasemia (HaT) than in comparable patients without a HaT diagnosis? A phenome-wide association study (PheWAS) is hypothesis-generating: it describes associations, not causes.

Data. Epic Cosmos (de-identified, multi-system US EHR), 2015-01-01 to 2026-06-01, ICD-10-CM diagnoses only, queried on a secured VM. Patient-level data never leave it.

Cases. Every patient with ICD-10-CM D89.44 (HaT); a single code suffices, because the diagnosis requires a genetic test (*TPSAB1* copy number). The index date is the first D89.44 not recorded as ruled out or entered in error.

Controls. A reproducible random pool of 300,000 Cosmos patients without D89.44, each indexed at one randomly chosen completed clinic visit from October 2021 (when D89.44 entered ICD-10-CM). Controls are chosen on their opportunity to be observed, never on their diagnoses.

Eligibility. At least 2 clinic-visit days in the year before index (both groups), a usable sex, and for cases an index from October 2021.

	HaT	Controls
In the pull	5,969	300,000
Eligible	4,144 (69%)	127,223 (42%)
Matched	3,911	28,670

Matching. Propensity-score nearest-neighbour matching (R MatchIt), up to 10 controls per case without replacement, exact on sex and calendar quarter of index, on age, years of record before and after index, prior-year clinic visits, race and ethnicity; caliper 0.2 SD. Every covariate balanced (standardized mean difference < 0.1) except follow-up after index (0.102), adjusted for after index. 5.6% of eligible cases found no match; mean 7.3 controls per case.

Outcomes and model. Diagnoses are grouped into phecodes (phecodeX). A patient is a phecode case with the code on 2 or more dates; sex-specific phecodes are analysed in one sex. For each phecode with at least 20 cases: logistic regression of the phecode on HaT, adjusted for age, sex, race, ethnicity, years of record and prior-year clinic visits (Firth regression when separation occurs),

with Bonferroni and false-discovery-rate (FDR) correction. **Primary window:** the 3 years before index. **Sensitivity window:** after index. The D89.44 code itself is removed from the outcomes. Every recorded ICD-10-CM diagnosis enters; about 82% of diagnosis rows map to a phecode, and the unmapped rows are encounter, status and screening Z-codes.

Verification. The published pyPheWAS package, run on identical inputs and compared under its own rules, agrees with our tool (pheauxWAS) phecode for phecode: correlation 0.999 where neither fit is separated, every estimate in the same direction.

2. What we found

	Before index (primary)	After index
Phecodes tested	1,817	1,611
Bonferroni-significant (higher in HaT)	482 (470)	514 (513)
FDR < 0.05 (higher in HaT)	858 (786)	865 (845)

The whole phenotype is shifted. The median odds ratio (OR) across all tested phecodes is 1.46 before index and 1.75 after, and ten negative-control phecodes with no known link to HaT (cataract, cerumen, myopia and others) give 1.42 and 1.89. HaT patients carry about 1.5–1.9 times the odds of almost any code, mostly from more contact with care: an ED visit in the year before index for 37% vs 30% of controls (not matched on), and 6.45 vs 4.23 clinic-visit days in the year after index. An OR of 3 is therefore only about twice the background; the leading phenotypes stand twenty to forty times above it.

Leading phenotypes (phecode cases on 2+ dates as a percentage of HaT patients and of controls; OR adjusted; all $q < 1e-130$):

Phenotype	HaT / controls, before	OR before	HaT / controls, after	OR after
Urticaria	25.6% / 0.7%	47.5	20.8% / 0.5%	55.1
↳ Idiopathic urticaria	7.2% / 0.1%	90.6	9.6% / 0.1%	123
Angioedema	6.4% / 0.1%	71.5	3.1% / 0.1%	60.5
Anaphylaxis	11.5% / 0.2%	61.2	8.8% / 0.2%	44.6
Flushing	8.5% / 0.7%	13.2	4.5% / 0.4%	9.3
Food allergy	13.0% / 1.2%	13.0	7.8% / 0.7%	11.8
“Allergy to insects” (mostly <i>allergy, unspecified</i>)	14.5% / 1.3%	13.1	7.5% / 0.8%	10.0
POTS	9.5% / 0.4%	30.4	10.1% / 0.3%	36.4
Hypermobility (includes Ehlers–Danlos codes)	9.9% / 0.3%	32.8	10.3% / 0.3%	39.9
Ehlers–Danlos	7.4% / 0.2%	40.8	8.1% / 0.1%	57.3
Rhinitis	36.1% / 13.4%	3.6	23.8% / 7.2%	3.9
Asthma	27.8% / 10.9%	3.1	22.9% / 7.2%	3.6
Abdominal pain	42.1% / 21.2%	2.6	30.8% / 12.1%	3.1

The cluster. Applying a stated rule ($FDR < 0.05$ in both windows and an OR at least twice that window's median) gives 156 phecodes in about 90 families. Beyond the table above they form: eosinophilic and functional gastrointestinal disease (eosinophilic esophagitis OR 10.5 / 13.9, small-intestinal bacterial overgrowth 12.2 / 13, gastroparesis, celiac disease, IBS); conditions reported alongside hypermobile EDS (Chiari-type malformations 11.2 / 18.1, CSF leak 7.0 / 15.8, celiac artery compression 7.5 / 12.5, orthostatic hypotension, chronic fatigue); immunoglobulin deficiencies (common variable immunodeficiency 18.4 / 19); eosinophilia (17.1 / 11.3); other genetic diagnoses that testing tends to find (alpha-1-antitrypsin deficiency 15.9 / 22.3, hemochromatosis); and a few unexpected rows, each in about 0.5% of HaT patients (some 20 people): liver malignancy and liver transplant, adrenal disorders, post-COVID condition. "Familial dysautonomia" (G90.1) in 5.2% of HaT patients is almost certainly the code used for dysautonomia in general.

Codes that belong to the diagnosis itself. The top hits by p-value include codes from the HaT workup, reported apart from the phenotype: mast cell activation codes D89.40–D89.49 (20.4% of HaT vs 0.0% of controls before index; often the label used before D89.44), R74.8 (the usual code for a raised tryptase; 40.4% vs 2.6%, falling after diagnosis), and mastocytosis codes D47.0x and C96.2x.

Mastocytosis subgroup. 30% of matched cases carry a mastocytosis code at some point, mostly D47.09 ("other mast cell neoplasms"); 9% carry systemic mastocytosis (D47.02). HaT is more common in systemic mastocytosis, and those patients are tested for it. The subgroup does not explain the phenotype: cases without these codes still show it (anaphylaxis 15.1% vs 0.6% of controls, POTS 10.2% vs 0.5%, hypermobility 10.5% vs 0.5%), and fractures show no excess. Whether D47.09 means confirmed clonal disease is a question for clinical review.

Tryptase. Only 18% of cases have a serum tryptase in the data, probably because it is run at outside laboratories; of those, 92% are at or above 8 ng/mL (median 15.3), as expected for HaT.

Lower in HaT. Tobacco use (OR 0.58), stimulant use disorders (0.20), atrial fibrillation (0.33), secondary malignancy (0.29), hypertension (0.80), diabetes (0.82 after index). These most plausibly reflect who is diagnosed with HaT (about 75% women, engaged in care) rather than protection.

3. How to read it

1. **The phenotype is consistent with HaT.** Mast-cell mediator symptoms, allergy, POTS and hypermobility lead in both windows, matching the multisystem picture described for HaT.
2. **Before diagnosis means "why patients were tested".** HaT is found by testing symptomatic patients, so the very large ORs before index describe the phenotype that leads to testing (indication), not consequences of HaT.
3. **Persistence after diagnosis shows the conditions are real and lasting, not that HaT causes them.** Chronic conditions keep being coded whatever their cause, and after diagnosis these patients are followed in clinics that look for POTS and EDS. Persistence rules out transient workup codes; it does not address indication.
4. **Much of the breadth is healthcare use.** The background shift (1.46–1.75) means modest associations should not be read as HaT-specific; only those well above it are.

4. Criticisms and limitations

Every weakness we can identify, each with its likely effect. **Major** marks those that could change the main conclusions.

Design and bias

1. **Indication bias (major).** Cases are tested because of their symptoms, so pre-index associations are inflated, likely heavily. It cannot be removed in this design; a genotyped population, or a comparator of patients tested for tryptase without HaT, would be needed.
2. **Asymmetric index events (major).** A case's index is a diagnostic, often specialist, visit; a control's is a random office visit, so the windows around them differ in intensity of evaluation. Remedy: anchor controls on a comparable evaluative visit, or lag the pre-index window by 6–12 months.
3. **Health-system and referral confounding (major).** HaT is diagnosed mainly at centres with mast-cell expertise; health system, region and payer are not matched. Remedy: match, stratify, or restrict controls to diagnosing systems.
4. **Residual healthcare intensity (major).** Matching on one year of office visits misses specialist, ED and testing intensity; the background shift and the ED excess above show it. Remedy: adjust for ED and specialist visits; calibrate on negative controls.
5. **Control misclassification.** Controls are not genotyped and HaT affects an estimated 4–6% of people, so some controls have it; bias toward the null, modest.
6. **Case misclassification.** Treated as negligible because a D89.44 requires a genetic test; ruled-out and error statuses are dropped. Genetic results are not in the data.
7. **Diagnosis-status filter.** The status values dropped (“rule”, “error”, “delete”, “cancel”) have not yet been checked against the values Cosmos actually uses.
8. **Selection by eligibility.** 29% of HaT patients fail the prior-utilization rule and 119 diagnosed before October 2021 are excluded; results describe HaT patients engaged in care before diagnosis.
9. **Code introduction.** D89.44 exists only from October 2021; 1,523 cases had an earlier D89.4x code, so the true diagnosis date may be earlier for some.
10. **Unmeasured population differences.** About 75% of cases are women; socioeconomic status, insurance and region are unmeasured, which likely explains the “lower in HaT” findings.
11. **Control-pool representativeness** against Cosmos as a whole has not been checked.
12. **Mastocytosis subgroup.** 30% of cases carry mastocytosis codes; the phenotype persists without them, so bias is modest. A sensitivity analysis excluding them is planned.

Matching

13. **The outcome model ignores the matched sets (major for inference).** Unconditional, unweighted regression; sets of 1 to 10 controls contribute unequally and within-set correlation is not modelled. Remedy: conditional logistic regression, or weighted regression with cluster-robust errors.
14. **Variable ratio.** 342 cases have 1 control, 2,210 have 10.
15. **Caliper scale.** 0.2 SD on the probability scale rather than the logit that Austin (2011) recommends.¹

¹Austin PC. Optimal caliper widths for propensity-score matching when estimating differences in means and differences in proportions in observational studies. *Pharmaceutical Statistics* 2011;10(2):150–161.

16. **Propensity model.** Main effects only; balance checked on means only.
17. **Follow-up after index** slightly imbalanced (0.102); adjusted in the post-index model.
18. **Pair-level differences** are large (e.g. age), though group means balance; relevant if a conditional analysis is adopted.

Outcomes

19. **Unequal observation time.** “Ever coded” in windows of varying length; time at risk is not modelled.
20. **Diagnosis-related codes (major for presentation).** Mast cell activation, raised tryptase and mastocytosis codes lead the results and must be reported apart from the phenotype.
21. **Phecode hierarchy.** Parent and child phecodes share patients; 482 or 858 significant phecodes are far fewer distinct findings.
22. **Case rule and controls.** Two dates are needed; phecodeX has no related-phecode exclusions, so controls can include related conditions (bias toward the null); denominators vary by phecode.
23. **Codes, not disease.** Billing codes reflect coding thoroughness, and phecode names can mislead (e.g. “allergy to insects” is mostly *allergy, unspecified*).

Statistics

24. **Odds ratios overstate risk ratios** for common outcomes (allergy affects about 21% of the cohort); the prevalence columns above show absolute differences.
25. **Firth regression for 21–24% of phecodes**, with Wald intervals and penalized likelihood-ratio p-values, which can disagree.
26. **Multiplicity under dependence.** Bonferroni is conservative for correlated phecodes; hierarchical FDR would fit better.
27. **Extreme p-values** underflow ($q < 1e-300$) and carry no ranking information; effect sizes should guide reading.
28. **Winner’s curse.** The largest effects will tend to shrink on replication.
29. **No replication** in an independent cohort yet.

Data and implementation

30. **Cosmos data quality.** Coding varies across organizations; site participation dates are not yet used to restrict observation.
31. **Pull window** starts in 2015, so a first D89.44 before then gets a later index (rare).
32. **Custom software.** pheauxWAS is purpose-written, verified against pyPheWAS and by self-tests. One fix during the work: sex-specific phecodes were first analysed in both sexes; the rerun with the fix changed little and all results here are from it.
33. **Small cells.** Counts of 1–10 must be masked before publication.

5. Planned sensitivity analyses

Analysis	Question
Remove all mast cell activation codes (and possibly R74.8 and mastocytosis) from outcomes	Does the phenotype stand without diagnosis-related coding?
Exclude cases with mastocytosis codes	Does the phenotype stand without them? (Early counts suggest yes.)
Conditional, or weighted cluster-robust, regression	Is inference robust to the matched design?
Remove controls with tryptase > 8 ng/mL	Effect of undiagnosed HaT among controls
Adjust for ED and specialist visits; negative-control calibration	How much of the breadth is healthcare use?
Match or stratify on health system or region	Is the breadth driven by where HaT is diagnosed?
Lag the pre-index window by 6–12 months	Are pre-index hits the workup just before diagnosis?
Require 3 full years before index; logit caliper; full 10:1 sets only	Sensitivity to window and matching choices
Cases with D89.44 on 2+ dates (optional)	Are single-code cases diluting or driving results?

6. Questions for reviewers

Clinical.

1. **Mastocytosis.** How many HaT patients in a practice like yours carry a mastocytosis diagnosis, and how reliable are D47.0x codes (is D47.09 used loosely)? Should the main phenotype be reported without them?
2. **Index date.** The first D89.44 is the anchor now. Would an earlier workup marker (tryptase, mast cell activation codes) be better?
3. **Diagnosis-related codes.** Which should be excluded in sensitivity analysis: D89.40–D89.49 only, or also R74.8 and mastocytosis codes?
4. Should controls with an elevated baseline tryptase be excluded?
5. **Ehlers–Danlos** is coded for 7.4% of HaT patients before diagnosis and 8.1% after. Does that fit clinical experience, and is hEDS usually diagnosed before or after HaT?
6. Do the unexpected cluster rows (liver malignancy and transplant, adrenal disorders, alpha-1-antitrypsin deficiency, post-COVID condition) fit your experience of HaT patients, or of the centres that diagnose them?

Statistical.

1. Is unconditional adjusted logistic regression acceptable as primary, given variable-ratio matching, or should conditional (or weighted cluster-robust) regression be primary?
2. Should the caliper move to the logit scale, as Austin recommends?
3. Multiplicity across dependent, nested phecodes: leaf level, collapsed to parents, or hierarchical FDR?

4. Which measures of healthcare intensity can be adjusted for without conditioning on outcomes?
5. How should extreme associations be presented when p underflows and Firth is used for a fifth of phecodes?
6. Health-system confounding: match, stratify, or restrict controls to diagnosing systems?
7. Time at risk: require full windows, or model rates?

Appendix: full findings

Every phecode in the cluster: FDR < 0.05 in both windows and an OR at least twice that window's median OR (2.92 before index, 3.51 after), with the diagnosis-related families (mast cell activation, raised tryptase, mastocytosis) left out. Children are listed under their parent; families are ordered by their strongest member. Percentages are phecode cases (2+ dates) among HaT patients and controls analysed for that phecode; female-only phecodes are percentages of women; <11 = 1–10 patients, masked.

Phecode	Description	OR pre [95% CI]	HaT / controls pre	OR post [95% CI]	HaT / controls post
DE_666	Urticaria	47.5 [40.5–55.6]	25.6% / 0.7%	55.1 [45.6–66.7]	20.8% / 0.5%
DE_666.2	↳ Idiopathic urticaria	90.6 [59.5–138]	7.2% / 0.1%	123 [80.7–189]	9.6% / 0.1%
DE_666.4	↳ Dermatographic urticaria	19 [10.8–33.4]	1.1% / 0.1%	56.4 [20.9–153]	0.7% / <11
DE_666.1	↳ Allergic urticaria	16.9 [7.58–37.5]	0.5% / <11	12.2 [5.54–26.9]	0.3% / <11
SS_812.2	Angioneurotic edema	71.5 [48.2–106]	6.4% / 0.1%	60.5 [35.3–104]	3.1% / 0.1%
SS_840	Allergy	7.95 [7.33–8.63]	63.9% / 19.0%	6.43 [5.9–7.02]	42.8% / 10.8%
SS_840.9	↳ Anaphylactic reaction	61.2 [46.5–80.5]	11.5% / 0.2%	44.6 [33.5–59.3]	8.8% / 0.2%
SS_840.1	↳ Food allergy	13 [11.2–15.1]	13.0% / 1.2%	11.8 [9.74–14.3]	7.8% / 0.7%
SS_840.18	↳ Egg allergy	10.9 [6.55–18.3]	0.8% / 0.1%	13.7 [7.59–24.8]	0.7% / 0.1%
SS_840.11	↳ Peanut allergy	8.07 [5.4–12.1]	1.3% / 0.2%	8.08 [4.94–13.2]	0.9% / 0.1%
SS_840.17	↳ Milk allergy	7.08 [4.63–10.8]	1.1% / 0.2%	8 [4.44–14.4]	0.6% / 0.1%
SS_840.12	↳ Seafood allergy	6.3 [4.33–9.17]	1.4% / 0.2%	7.4 [4.72–11.6]	1.1% / 0.1%
SS_840.2	↳ Allergy to insects	13.1 [11.4–15.2]	14.5% / 1.3%	9.97 [8.28–12]	7.5% / 0.8%
SS_840.6	↳ Drug and medical agent allergy	3.37 [3.04–3.75]	18.5% / 6.7%	3.73 [3.3–4.21]	13.2% / 3.9%
SS_840.62	↳ Allergy to anesthetic agent	5.74 [2.67–12.3]	0.3% / 0.1%	10.6 [4.94–22.8]	0.4% / <11
SS_840.64	↳ Allergy to analgesic agent	3.88 [3.03–4.98]	2.6% / 0.7%	4.63 [3.52–6.09]	2.4% / 0.5%
SS_840.65	↳ Allergy to serum and vaccine	3.45 [1.93–6.16]	0.4% / 0.1%	5.74 [3.06–10.7]	0.4% / 0.1%

Phecode	Description	OR pre [95% CI]	HaT / controls pre	OR post [95% CI]	HaT / controls post
SS_840.5	↳ Allergy to radiographic dye	5.55 [3.95–7.78]	1.5% / 0.3%	6.78 [4.83–9.53]	1.7% / 0.3%
SS_840.4	↳ Latex allergy	4.3 [3.25–5.7]	2.1% / 0.5%	5.32 [3.81–7.42]	1.7% / 0.3%
SS_840.8	↳ Allergies related to other diseases/symptoms	3.98 [3.64–4.35]	28.4% / 9.0%	4.01 [3.6–4.47]	18.1% / 4.9%
GE_972	Genetic neuromuscular disease	8.05 [6.67–9.71]	6.2% / 0.8%	11.4 [9.29–13.9]	6.7% / 0.6%
GE_972.5	↳ Familial dysautonomia [Riley-Day] (G90.1)	43.6 [30.3–62.8]	5.2% / 0.1%	46.9 [32.3–68]	5.6% / 0.1%
GE_978	Genetic disorders of growth and musculoskeletal system	27.1 [21.1–34.8]	7.6% / 0.3%	38.4 [29.1–50.8]	8.3% / 0.2%
GE_978.2	↳ Hereditary connective tissue diseases	37.5 [28.2–49.8]	7.4% / 0.2%	56.6 [40.8–78.6]	8.2% / 0.1%
GE_978.22	↳ Ehlers–Danlos syndrome	40.8 [30.4–54.8]	7.4% / 0.2%	57.3 [41.1–79.9]	8.1% / 0.1%
MS_712	Joint derangements and related disorders	8.5 [7.43–9.73]	13.0% / 1.7%	13.3 [11.4–15.5]	13.0% / 1.0%
MS_712.5	↳ Disorder of ligament	29 [23.3–36.1]	10.0% / 0.4%	38.7 [30.4–49.4]	10.5% / 0.3%
MS_712.51	↳ Hypermobility syndrome	32.8 [26.1–41.4]	9.9% / 0.3%	39.9 [31.2–51.2]	10.3% / 0.3%
MS_712.63	↳ Spinal instabilities	6.69 [3.85–11.6]	0.6% / 0.1%	10.2 [5.64–18.3]	0.7% / 0.1%
MS_712.3	↳ Articular cartilage disorder	4.33 [2.35–7.97]	0.4% / 0.1%	4.23 [2.22–8.03]	0.3% / 0.1%
NS_343	Disorders of autonomic nervous system	22.2 [18.6–26.5]	12.8% / 0.6%	24.3 [20.1–29.2]	12.9% / 0.6%
NS_343.7	↳ Postural orthostatic tachycardia syndrome	30.4 [24.2–38.1]	9.5% / 0.4%	36.4 [28.6–46.4]	10.1% / 0.3%
NS_343.3	↳ Complex regional pain syndrome	5.42 [3.43–8.56]	0.8% / 0.1%	5.1 [3.16–8.23]	0.8% / 0.1%
SS_841	Adverse effect of food (not allergy)	27.1 [19.9–36.8]	4.9% / 0.2%	37.7 [25–56.9]	3.6% / 0.1%
BI_179	Immunodeficiencies	5.09 [4.34–5.98]	7.1% / 1.5%	6.29 [5.36–7.38]	8.0% / 1.3%
BI_179.7	↳ Common variable immunodeficiency	18.4 [10–33.5]	1.0% / 0.1%	19 [10.7–33.7]	1.2% / 0.1%
BI_179.6	↳ Selective deficiency of IgG subclasses	12.7 [6.56–24.7]	0.6% / 0.0%	12.6 [6.63–23.9]	0.6% / 0.0%

Phecode	Description	OR pre [95% CI]	HaT / controls pre	OR post [95% CI]	HaT / controls post
BI_179.1	↳ Hypogammaglobulinemia NOS	10.3 [7.15–14.7]	1.9% / 0.2%	10.3 [7.2–14.8]	2.0% / 0.2%
BI_179.4	↳ Selective IgA deficiency	8.75 [5–15.3]	0.6% / 0.1%	13.2 [6.1–28.5]	0.4% / <11
BI_179.9	↳ Immunodeficiency NOS	4.23 [3.38–5.28]	3.4% / 0.8%	5.29 [4.24–6.61]	3.8% / 0.7%
GE_981	Other genetic diseases (arbitrary group 1)	9.03 [4.99–16.3]	0.5% / 0.1%	16.7 [9.07–30.6]	0.9% / 0.1%
GE_981.1	↳ Alpha-1-antitrypsin deficiency	15.9 [8.05–31.6]	0.5% / <11	22.3 [11.7–42.7]	0.8% / <11
GI_522.5	Allergic and dietetic gastroenteritis and colitis	14.9 [7.42–29.8]	0.5% / 0.0%	22.4 [9.4–53.6]	0.4% / <11
GI_532	Other disorders of the intestines	3.27 [2.65–4.02]	3.6% / 1.1%	3.95 [3.11–5.01]	2.9% / 0.7%
GI_532.5	↳ Small intestinal bacterial overgrowth	12.2 [8.03–18.5]	1.6% / 0.1%	13 [8.07–20.9]	1.4% / 0.1%
BI_171.7	Eosinophilia	17.1 [12–24.5]	2.5% / 0.2%	11.3 [7.14–17.9]	1.2% / 0.1%
MS_700	Diffuse diseases of connective tissue	4.2 [3.59–4.92]	6.9% / 1.7%	4.77 [4.02–5.67]	6.3% / 1.2%
MS_700.6	↳ Autoinflammatory syndromes	11.2 [5.99–21.1]	0.6% / 0.0%	30.3 [13.1–70]	0.6% / <11
MS_700.2	↳ Sicca syndrome [Sjogren syndrome]	4.79 [3.68–6.23]	2.5% / 0.5%	6.06 [4.6–7.97]	2.7% / 0.4%
CM_750	Congenital malformations of nervous system	11.2 [9.05–13.8]	5.8% / 0.6%	18.1 [14.2–23.1]	6.2% / 0.3%
GI_522.6	Eosinophilic gastroenteritis and colitis	11 [8.01–15.1]	2.5% / 0.2%	15.1 [10.9–20.9]	3.0% / 0.2%
GI_522.61	↳ Eosinophilic esophagitis	10.5 [7.6–14.6]	2.3% / 0.2%	13.9 [9.96–19.5]	2.7% / 0.2%
DE_679	Skin symptoms	6.86 [6.25–7.52]	31.4% / 6.3%	5.11 [4.55–5.75]	16.3% / 3.5%
DE_679.3	↳ Flushing	13.2 [11–15.9]	8.5% / 0.7%	9.26 [7.3–11.7]	4.5% / 0.4%
DE_679.4	↳ Pruritus	7.11 [6.03–8.39]	7.9% / 1.2%	6.11 [4.96–7.53]	4.7% / 0.8%
DE_679.1	↳ Rash and other nonspecific skin eruption	5.25 [4.65–5.91]	14.7% / 3.1%	3.52 [2.98–4.16]	6.3% / 1.7%
DE_668.6	Prurigo	7.64 [4.51–13]	0.7% / 0.1%	8.99 [4.99–16.2]	0.6% / 0.1%

Phecode	Description	OR pre [95% CI]	HaT / controls pre	OR post [95% CI]	HaT / controls post
CV_443.2	Celiac artery compression syndrome	7.45 [4.05–13.7]	0.5% / 0.1%	12.5 [6.17–25.4]	0.5% / <11
SS_847.3	Liver transplant	7.45 [4.31–12.9]	0.6% / 0.1%	7.76 [4.57–13.2]	0.7% / 0.1%
EM_211	Disorders of adrenal glands	3.66 [2.94–4.56]	3.3% / 0.9%	4.15 [3.26–5.29]	2.9% / 0.6%
EM_211.1	↳ Adrenal hyperfunction	5.35 [3.45–8.3]	0.9% / 0.2%	5.96 [3.49–10.2]	0.7% / 0.1%
EM_211.13	↳ Cushing's syndrome	7.44 [3.96–14]	0.5% / 0.1%	7.76 [3.74–16.1]	0.4% / <11
EM_211.11	↳ Hyperaldosteronism	5.61 [2.92–10.8]	0.3% / 0.1%	4.35 [2.11–8.98]	<11 / 0.1%
EM_211.2	↳ Adrenocortical insufficiency [Addison's disease]	5.82 [4.23–8.01]	1.8% / 0.3%	6.22 [4.54–8.52]	2.0% / 0.3%
RE_463	Rhinitis and nasal congestion	3.59 [3.31–3.9]	36.1% / 13.4%	3.85 [3.49–4.25]	23.8% / 7.2%
RE_463.1	↳ Chronic rhinitis	7.07 [5.91–8.45]	6.6% / 1.0%	9.86 [7.87–12.4]	5.1% / 0.5%
RE_463.2	↳ Allergic rhinitis	3.53 [3.21–3.87]	23.7% / 8.0%	3.54 [3.16–3.97]	14.9% / 4.4%
RE_463.23	↳ Allergic rhinitis, due to animal hair and dander	6.59 [4.67–9.31]	1.6% / 0.2%	9.38 [6.25–14.1]	1.5% / 0.1%
RE_463.3	↳ Vasomotor rhinitis	4.89 [2.71–8.84]	0.5% / 0.1%	6.43 [3.77–11]	0.6% / 0.1%
NS_344	Disorders of the circulation of the cerebrospinal fluid	3.18 [2.31–4.39]	1.4% / 0.4%	3.58 [2.58–4.97]	1.5% / 0.4%
NS_344.3	↳ Cerebrospinal fluid leak	7.04 [3.82–13]	0.5% / 0.1%	15.8 [7.37–33.8]	0.5% / <11
NS_344.2	↳ Benign intracranial hypertension	3.7 [2.33–5.89]	0.7% / 0.2%	4.31 [2.77–6.71]	0.9% / 0.2%
GE_962	Disorders of amino-acid transport and metabolism	3.15 [1.91–5.22]	0.6% / 0.2%	4.07 [2.42–6.83]	0.6% / 0.1%
GE_962.4	↳ Disturbances of sulphur-bearing amino-acid metabolism	6.65 [3.27–13.5]	0.4% / 0.1%	8.16 [4.25–15.7]	0.5% / 0.0%
SS_807.3	Post COVID-19 condition	6.4 [4.82–8.49]	2.4% / 0.4%	12.2 [8.52–17.6]	2.2% / 0.2%
ID_020	Borrelia	6.13 [3.75–10]	0.8% / 0.1%	6.41 [3.62–11.4]	0.5% / 0.1%
ID_020.1	↳ Lyme disease	4.55 [2.63–7.87]	0.5% / 0.1%	4.88 [2.5–9.54]	0.3% / 0.1%

Phecode	Description	OR pre [95% CI]	HaT / controls pre	OR post [95% CI]	HaT / controls post
GU_625.3	Vulvodynia	6 [3.22–11.2]	0.6% / 0.1%	7.7 [3.69–16]	0.5% / <11
CA_101.6	Malignant neoplasm of the liver and intrahepatic bile ducts	5.08 [2.86–9.03]	0.5% / 0.1%	4.35 [2.45–7.75]	0.5% / 0.1%
CA_101.61	↳ Malignant neoplasm of the liver	6.74 [3.61–12.6]	0.5% / 0.1%	5.93 [3.17–11.1]	0.4% / 0.1%
DE_672.1	Acute skin changes due to ultraviolet radiation	6.26 [3.56–11]	0.6% / 0.1%	5.9 [2.74–12.7]	0.3% / 0.0%
GI_525	Intestinal malabsorption	5.83 [4.9–6.93]	6.5% / 1.1%	6.93 [5.65–8.49]	5.2% / 0.7%
GI_525.1	↳ Celiac disease	5.35 [4–7.15]	2.1% / 0.4%	6.24 [4.46–8.73]	1.8% / 0.3%
DE_668.4	Dermatitis due to substances taken internally	5.69 [3.69–8.77]	1.0% / 0.2%	5.71 [3.16–10.3]	0.5% / 0.1%
NS_333.7	Narcolepsy	5.52 [3.57–8.53]	0.9% / 0.1%	6.88 [4.17–11.3]	0.8% / 0.1%
SS_807.1	Chronic fatigue syndrome	5.46 [4.71–6.32]	9.0% / 1.7%	6.53 [5.45–7.83]	6.5% / 0.9%
RE_472	Diseases of vocal cords and larynx, not elsewhere classified	5.32 [4.15–6.81]	2.9% / 0.5%	5.65 [4.2–7.59]	2.2% / 0.4%
RE_472.4	↳ Laryngeal spasm	7.06 [3.53–14.1]	0.3% / 0.0%	4.97 [2.33–10.6]	<11 / 0.0%
GI_516	Other diseases of stomach and duodenum	3.95 [3.31–4.72]	5.5% / 1.5%	5.56 [4.61–6.72]	5.6% / 0.9%
GI_516.4	↳ Gastroparesis	5.3 [4.17–6.75]	3.2% / 0.6%	7.25 [5.67–9.28]	3.7% / 0.4%
GI_510.5	Dyskinesia of esophagus	5.07 [3.36–7.63]	1.0% / 0.2%	6.63 [4.27–10.3]	1.1% / 0.2%
SO_367.12	Allergic conjunctivitis	4.88 [3.74–6.37]	2.4% / 0.5%	7.97 [5.68–11.2]	2.0% / 0.2%
GI_552.1	Cholangitis	5.31 [2.76–10.2]	0.4% / 0.1%	4.74 [2.44–9.21]	0.4% / 0.1%
CV_446.2	Orthostatic hypotension	5.12 [4.15–6.31]	4.1% / 0.8%	4.7 [3.69–5.97]	3.1% / 0.7%
SS_826	Other abnormal immunological findings in serum	4.27 [3.68–4.95]	8.0% / 1.9%	5.03 [4.18–6.05]	5.5% / 1.0%
SS_826.4	↳ Other and unspecified nonspecific immunological findings	4.66 [3.99–5.45]	7.4% / 1.6%	5.14 [4.21–6.27]	4.7% / 0.9%

Phecode	Description	OR pre [95% CI]	HaT / controls pre	OR post [95% CI]	HaT / controls post
NS_335	Nerve root and plexus disorders	3.19 [2.13–4.77]	0.9% / 0.3%	4.63 [3.05–7.02]	1.0% / 0.2%
NS_335.1	↳ Nerve plexus lesions	4.64 [2.9–7.43]	0.7% / 0.2%	6.08 [3.65–10.1]	0.8% / 0.1%
NS_324.33	Fasciculation	9.22 [5.4–15.7]	0.8% / 0.1%	4.48 [2.27–8.84]	0.4% / 0.1%
GI_529.2	Flatulence, eructation, and gas pain	4.36 [3.78–5.02]	8.8% / 2.1%	4.37 [3.63–5.25]	5.3% / 1.1%
CV_448.1	Raynaud's syndrome	4.33 [3.37–5.57]	2.7% / 0.6%	4.92 [3.71–6.52]	2.3% / 0.4%
NS_334.1	Trigeminal nerve disorders [CN5]	4.3 [3.04–6.09]	1.3% / 0.3%	4.43 [2.94–6.66]	1.0% / 0.2%
NS_334.11	↳ Trigeminal neuralgia	4.18 [2.75–6.35]	0.9% / 0.2%	3.77 [2.37–6.01]	0.7% / 0.2%
DE_685	Disorders of sweat glands	4.01 [3.2–5.02]	3.3% / 0.8%	4.08 [3.07–5.41]	2.1% / 0.5%
DE_685.8	↳ Hyperhidrosis	4.27 [3.33–5.48]	2.7% / 0.6%	4.09 [2.99–5.61]	1.7% / 0.4%
DE_685.82	↳ Generalized hyperhidrosis	4.53 [3.52–5.85]	2.7% / 0.6%	4.24 [3.08–5.84]	1.7% / 0.4%
EM_230.5	Early satiety	4.19 [2.99–5.86]	1.4% / 0.3%	4.64 [3.07–7]	1.1% / 0.2%
RE_481.1	Pulmonary eosinophilia	4.44 [2.17–9.1]	<11 / 0.1%	4.17 [1.93–9]	<11 / 0.0%
GI_524	Functional intestinal disorder	4.16 [3.66–4.73]	11.1% / 2.8%	4.68 [4.02–5.45]	8.4% / 1.7%
GI_524.1	↳ Irritable bowel syndrome	3.91 [3.42–4.47]	9.9% / 2.6%	4.45 [3.8–5.22]	7.6% / 1.6%
SS_835	Cytology and pathology findings	4.16 [3.17–5.45]	2.3% / 0.5%	4.95 [3.55–6.92]	1.6% / 0.3%
RE_471.5	Nasal polyps	4.06 [2.6–6.35]	0.8% / 0.2%	7.04 [4.36–11.4]	0.9% / 0.1%
MS_717.3	Occipital neuralgia	3.96 [2.85–5.5]	1.5% / 0.4%	5.33 [3.74–7.59]	1.5% / 0.2%
GI_527.3	Functional dyspepsia	3.92 [2.75–5.59]	1.3% / 0.3%	6.46 [4.21–9.92]	1.1% / 0.2%
GI_506.5	Disturbances of salivary secretion	3.82 [2.73–5.34]	1.4% / 0.4%	4.27 [2.91–6.27]	1.1% / 0.2%
GI_506.51	↳ Dry mouth	3.92 [2.6–5.9]	0.9% / 0.2%	5.4 [3.43–8.53]	0.8% / 0.1%
NS_337.2	Inflammatory polyneuropathy	3.53 [1.98–6.29]	0.4% / 0.1%	3.9 [2.07–7.35]	0.4% / 0.1%
NS_337.22	↳ Chronic inflammatory demyelinating polyneuritis	3.83 [1.72–8.55]	<11 / 0.1%	5.98 [2.49–14.4]	<11 / 0.0%

Phecode	Description	OR pre [95% CI]	HaT / controls pre	OR post [95% CI]	HaT / controls post
GI_522.8	Duodenitis	3.83 [2.25–6.5]	0.5% / 0.2%	4.98 [2.74–9.06]	0.4% / 0.1%
NS_321	Encephalitis, myelitis and encephalomyelitis	3.82 [2.04–7.15]	0.3% / 0.1%	4.79 [2.49–9.2]	0.3% / 0.1%
GI_507.11	Recurrent oral aphthae [Recurrent aphthous stomatitis]	7.29 [3.97–13.4]	0.5% / 0.1%	3.79 [1.84–7.81]	<11 / 0.1%
MS_724.1	Myalgia	3.77 [3.39–4.2]	17.3% / 5.3%	4.21 [3.73–4.75]	13.9% / 3.4%
GU_592.12	Chronic cystitis	3.73 [2.71–5.14]	1.5% / 0.4%	4.74 [3.22–6.98]	1.2% / 0.2%
ID_052.4	Infectious mononucleosis	3.65 [2.15–6.22]	0.5% / 0.1%	6.82 [3.17–14.7]	0.4% / 0.0%
NS_350.3	Abnormal reflex	3.63 [2.18–6.05]	0.6% / 0.2%	4.2 [2.17–8.12]	0.4% / 0.1%
RE_494	Voice disturbance	3.34 [2.63–4.23]	2.7% / 0.9%	4.29 [3.29–5.6]	2.4% / 0.5%
RE_494.1	↳ Dysphonia	3.6 [2.79–4.64]	2.4% / 0.7%	4.72 [3.58–6.22]	2.3% / 0.5%
MB_291	Dissociative and somatoform disorders	3.17 [2.34–4.31]	1.6% / 0.5%	3.86 [2.79–5.33]	1.6% / 0.4%
MB_291.1	↳ Conversion disorder	3.53 [2.2–5.66]	0.7% / 0.2%	4.62 [2.98–7.16]	0.9% / 0.2%
NS_331.63	Menstrual migraine	4.11 [2.23–7.56]	0.6% / 0.1%	3.51 [1.83–6.72]	0.4% / 0.1%
DE_668.3	Contact dermatitis	3.48 [2.78–4.35]	3.1% / 0.9%	4.89 [3.74–6.4]	2.6% / 0.5%
NS_331.61	Migraine with aura	3.48 [2.95–4.1]	6.0% / 1.7%	3.89 [3.24–4.67]	5.3% / 1.2%
GI_522.14	Microscopic colitis	3.47 [2.22–5.43]	0.7% / 0.2%	4.52 [2.77–7.36]	0.6% / 0.1%
NS_331.64	Chronic migraine without aura	3.45 [2.91–4.09]	5.6% / 1.5%	3.68 [3.07–4.41]	5.3% / 1.3%
EM_247.71	Hemochromatosis	3.44 [2.1–5.64]	0.6% / 0.2%	3.99 [2.38–6.68]	0.6% / 0.2%
EM_252.51	Disorders of intestinal carbohydrate absorption	3.41 [2.2–5.29]	0.8% / 0.2%	4.38 [2.53–7.57]	0.6% / 0.1%
MS_741.4	Sprains and strains of hip and thigh	3.41 [2.29–5.09]	1.0% / 0.3%	4.87 [2.9–8.17]	0.7% / 0.1%
BI_168.3	Spontaneous ecchymoses	3.38 [2.29–4.98]	1.0% / 0.3%	3.76 [2.28–6.2]	0.6% / 0.1%
ID_052.5	Cytomegalovirus [CMV]	3.34 [1.73–6.44]	0.3% / 0.1%	8.58 [4.12–17.9]	0.3% / 0.0%

Phecode	Description	OR pre [95% CI]	HaT / controls pre	OR post [95% CI]	HaT / controls post
NS_331.3	Headache syndromes, non migraine	3.32 [2.66–4.16]	3.1% / 0.9%	3.92 [3.02–5.09]	2.5% / 0.5%
GI_554.3	Exocrine pancreatic insufficiency	3.32 [1.67–6.58]	0.3% / 0.1%	6.78 [3.56–12.9]	0.4% / 0.1%
GE_967	Genetic disorders of vitamin and mineral metabolism	3.07 [1.74–5.41]	0.4% / 0.2%	5.14 [2.98–8.87]	0.6% / 0.1%
GE_967.4	↳ Hereditary hemochromatosis	3.3 [1.8–6.07]	0.4% / 0.1%	4.01 [2.26–7.1]	0.4% / 0.1%
NS_331.4	Cluster headaches	3.27 [1.86–5.72]	0.5% / 0.1%	5.62 [2.68–11.8]	0.3% / 0.1%
SS_829.2	Abnormal level of blood mineral	3.26 [2.34–4.53]	1.4% / 0.4%	3.66 [2.57–5.22]	1.3% / 0.3%
GE_982	Genetic susceptibility of disease, NOS	3.23 [2.46–4.23]	2.1% / 0.6%	4.33 [3.26–5.74]	2.2% / 0.4%
RE_477	Inhalation lung injury	3.2 [1.62–6.29]	<11 / 0.1%	3.79 [1.78–8.08]	<11 / 0.1%
GU_615	Endometriosis	3.12 [2.4–4.07]	2.8% / 0.9%	3.77 [2.76–5.16]	2.2% / 0.6%
CM_768	Congenital deformities of chest and bony thorax	3.16 [1.61–6.2]	<11 / 0.1%	4.3 [1.97–9.4]	<11 / 0.0%
CV_417.1	Palpitations	3.13 [2.82–3.48]	15.9% / 5.6%	3.58 [3.14–4.08]	10.4% / 2.9%
SO_395	Other diseases of inner ear	3.07 [1.54–6.08]	<11 / 0.1%	3.92 [1.89–8.14]	<11 / 0.1%
RE_475	Asthma	3.06 [2.82–3.33]	27.8% / 10.9%	3.59 [3.26–3.94]	22.9% / 7.2%
DE_668.1	Atopic dermatitis	3.05 [2.49–3.74]	3.7% / 1.4%	4.55 [3.59–5.77]	3.1% / 0.7%
DE_668.5	Lichen simplex chronicus	3.02 [1.57–5.82]	0.3% / 0.1%	3.59 [1.59–8.07]	<11 / 0.0%
NS_333.4	Circadian rhythm sleep disorder	2.98 [1.86–4.78]	0.6% / 0.2%	4.28 [2.39–7.67]	0.4% / 0.1%
BI_174.2	Splenomegaly	2.96 [2.12–4.14]	1.3% / 0.4%	5.77 [3.77–8.84]	1.0% / 0.2%